Lab 7: Fuzzing for Input Validation Bugs

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Repository: L6_L7_31_coupon_validator (lab7)

Step 2: Running the buggy application (python coupon_validator.py)

The original script works for valid input, but crashes with a ValueError when the coupon code has no dash (input: save).

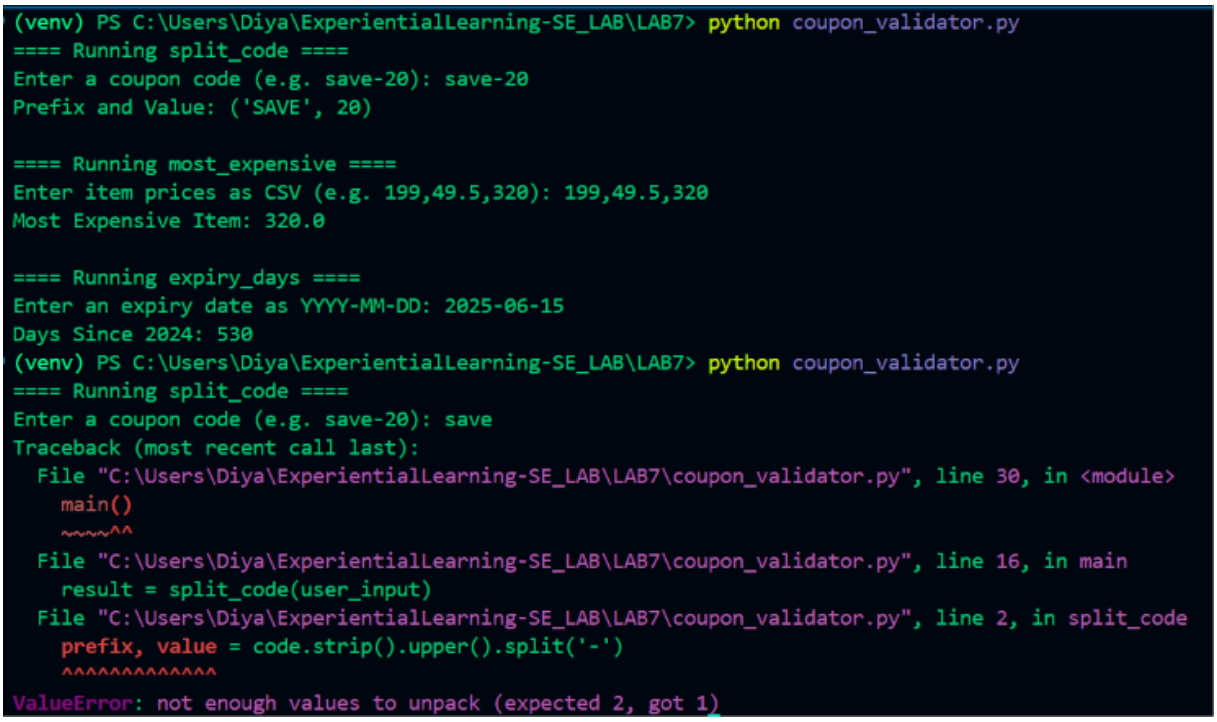


Figure 1: Valid input works. Input 'save' crashes in split_code (line 2).

Step 4: Running the Hypothesis tests against the buggy code

Hypothesis generated random inputs and found failures in all three functions. The failing test case for split_code includes s='-', s='' and s=None, shown below.

```
(venv) PS C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7> python -m pytest
========================================= test session starts =========================================
platform win32 -- Python 3.14.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7
plugins: hypothesis-6.170.15, platformdirs-4.13.7
collected 9 items

test_coupon_validator.py FFFF..F.F                                                          [100%]

============================================== FAILURES ===============================================
_____________________________________ test_split_code_no_crash ______________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\test_coupon_validator.py", line 11, in test_spl
it_code_no_crash
  |     def test_split_code_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\test_coupon_validator.py", line 13, in test_s
plit_code_no_crash
    |     split_code(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\coupon_validator.py", line 3, in split_code
    |     return prefix, int(value)
```

Figure 2: pytest starts: 9 tests collected, the first tests fail.

```
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\test_coupon_validator.py", line 15, in test_s
plit_code_no_crash
    |     assert False, f"split_code crashed with: {e}"
    | AssertionError: split_code crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_split_code_no_crash(
    |     s='-',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\test_coupon_validator.py", line 13, in test_s
plit_code_no_crash
    |     split_code(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\coupon_validator.py", line 2, in split_code
    |     prefix, value = code.strip().upper().split('-')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
```

Figure 3: Failing test case 1 for split_code: s='-' (int() on an empty value).

```
    | Traceback (most recent call last):
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\test_coupon_validator.py", line 15, in test_s
plit_code_no_crash
    |     assert False, f"split_code crashed with: {e}"
    | AssertionError: split_code crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_split_code_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\test_coupon_validator.py", line 13, in test_s
plit_code_no_crash
    |     split_code(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\coupon_validator.py", line 2, in split_code
    |     prefix, value = code.strip().upper().split('-')
    |                     ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7\test_coupon_validator.py", line 15, in test_s
plit_code_no_crash
    |     assert False, f"split_code crashed with: {e}"
```

Figure 4: Failing test case 2 for split_code: s='' (not enough values to unpack), and failing case 3 starting.

```
    | AssertionError: split_code crashed with: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_split_code_no_crash(
    |     s=None,  # or any other generated value
    | )
    +-----------------------------------
```

Figure 5: Failing test case 3 for split_code: s=None (NoneType has no attribute 'strip').

```
========================================= short test summary info =========================================
FAILED test_coupon_validator.py::test_split_code_no_crash - ExceptionGroup: Hypothesis found 3 distinct fa
ilures. (3 sub-exceptions)
FAILED test_coupon_validator.py::test_most_expensive_no_crash - ExceptionGroup: Hypothesis found 2 distinc
t failures. (2 sub-exceptions)
FAILED test_coupon_validator.py::test_expiry_days_no_crash - ExceptionGroup: Hypothesis found 2 distinct f
ailures. (2 sub-exceptions)
FAILED test_coupon_validator.py::test_non_string_inputs_are_rejected_safely - ExceptionGroup: Hypothesis f
ound 2 distinct failures. (2 sub-exceptions)
FAILED test_coupon_validator.py::test_most_expensive_never_returns_nan_or_inf - ExceptionGroup: Hypothesis
 found 2 distinct failures. (2 sub-exceptions)
FAILED test_coupon_validator.py::test_expiry_days_returns_int_or_none - ExceptionGroup: Hypothesis found 2
 distinct failures. (2 sub-exceptions)
====================================== 6 failed, 3 passed in 15.04s =======================================
```

Figure 6: Summary with the buggy code: 6 failed, 3 passed.

Step 5: Running the fixed application (python coupon_validator.py)

After hardening the three functions, invalid input now gives a clear message instead of a crash, and valid input gives the same results as before.

```
(venv) PS C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7> python coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): save
Invalid coupon code. Use the form PREFIX-NUMBER.

==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): 199,49.5,320
Most Expensive Item: 320.0

==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: 2025-06-15
Days Since 2024: 530
(venv) PS C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7> python coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): save-20
Prefix and Value: ('SAVE', 20)

==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): 199,49.5,320
Most Expensive Item: 320.0

==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: 2025-06-15
Days Since 2024: 530
```

Figure 7: Input 'save' now prints an invalid code message; valid input still works.

Step 6: Re-running the tests against the fixed code

All Hypothesis tests now pass.

```
(venv) PS C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7> python -m pytest test_coupon_validator.py
========================================= test session starts =========================================
platform win32 -- Python 3.14.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Diya\ExperientialLearning-SE_LAB\LAB7
plugins: hypothesis-6.170.15, platformdirs-4.13.7
collected 9 items

test_coupon_validator.py .........                                                          [100%]

========================================== 9 passed in 2.49s ==========================================
```

Figure 8: python -m pytest test_coupon_validator.py: 9 passed.